

AI Project Overview

The purpose of this project is to gain a deeper understanding of how a program can take human speech input and figure out/ complete the task given to it. While many such programs already exist, the goal is to use as few API's/ imports as possible, instead trying to build up and be a part of the machine learning process to help gain the most knowledge on how it works.

My Project will have been a success if I can:

- Understand and explain the data pipeline that a voice-based assistant would use
 - I want to be able to accurately explain how audio input can be taken from a microphone, turned into a request that can be used by the assistant, and then how the assistant can form a verbal response to it that is not just a preset sentence. Essentially I wish to be able to explain how to make an AI “think” and respond like a human would and not just respond from a list of sentences.
- Develop a speech-to-text system with little reliance on external APIs
 - While using APIs will make this project more simple, I will not be able to learn anything from the process. I wish to be able to see how speech recognition works and implement as much of the audio-processing pipeline myself as possible. Any APIs used throughout the project, whether for audio or not, will be documented and explained as to why they are used.
- Develop a way to determine if the user is speaking to the assistant or just speaking
 - There should be a specific trigger word that will cause the AI to actually start “listening” and receiving the input the user wants. This will help me learn how to prevent the AI from continuously taking in tasks and storing data that does not need to be stored.
- Use the AI's interpretation of the tasks to actually complete them
 - I will have a task system that the input will be sent to once the task is interpreted that will actually complete the task.
- Develop a way to handle interruptions
 - The assistant should be able to pause / restart if the user begins speaking while the AI is. This will help me learn how to make an AI able to process multiple things at once instead of being stuck to whatever it is that it is currently doing
- Implement a basic form of memory
 - As far as I am aware, there are not any classes that teach how AI can remember what the user asks it or what its response is. I want to be able to understand and implement a way for these things to be stored and retrieved if asked during a future interaction

Project Features:

- Listen for its name to be called before preparing to take in a task from the user
 - This will show my understanding of how the AI can constantly “listen” for its keyword, while not actually storing any data or completing any tasks
- Display the words the user is saying as they speak them
 - This will help to demonstrate my understanding of the speech-to-text pipeline

- Allow and prepare for interruptions from user if program is “speaking”
- Respond with a “voice” to explain the answer
 - While it just says responding with a voice, this feature will show my understanding of how an AI can “think” and respond like a person would
- Display the local weather when asked (temperature, chance of rain, highs and lows) depending on what they ask (if just asked for weather will only say chance of rain and temperature)
- Displaying local restaurants / businesses (Ex: “What is a good mexican restaurant near me?”)
- Set / remove timers
- Simple Google searches
- Access / play Spotify music
- Add / remove specific reminders onto a calender
- Create notes and move them to specific folders if asked
- Have multiple voices that the user can choose from
- Adjust its own volume if prompted to
 - These tasks will show my knowledge of turning the user’s request into a usable format and how it is used to complete the task
- Display an audio visualizer while talking
 - This will further prove my understanding of audio processing
- Have some memory of previous inquires / preferences
 - This will prove that I gained an understanding of how memory works in regards to AI

Once again the purpose here is to learn and understand how AI can interact with humans in a way that feels more natural than just bringing up a google search, without just importing the latest model of ChatGPT into the program. I want to be a part of the learning process and really understand how AI can be trained to respond “human” like while also being able to complete the tasks it is given.